

WTW 158 Worksheet 5 on Limit laws and Continuity

A. TEXTBOOK PROBLEMS

p 102 no 21, 23, 39, 45, 49

p196 no 39, 45, 50

p124 no 19, 23, 41, 45

B. MORE PROBLEMS

1. Given the function $f(x) = \begin{cases} e^{-x+1} & \text{if } x < 1 \\ k & \text{if } x = 1 \\ \frac{2}{x} & \text{if } x > 1 \end{cases}$

Find k (if possible) such that

(i) f is left continuous in $x = 1$ (ii) f is right continuous in $x = 1$ (iii) f is continuous in $x = 1$

2. $f(x) = \begin{cases} \frac{\cos x - 1}{\sin x} & \text{if } x < 0 \\ 1 & \text{if } x = 0 \\ \frac{1}{\ln(x+1)} & \text{if } x > 0 \end{cases}$

Investigate the continuity of f in $x = 0$.

3. State the domain on which the following functions are continuous.

(i) $y = \arcsin x$ (ii) $f(x) = \frac{\tan x}{\sqrt{1-x^2}}$

C. TEST QUESTIONS

1. Find the limit (if it exists):

1.1 $\lim_{x \rightarrow 2} \sec \frac{\pi x}{3}$ 1.2 $\lim_{x \rightarrow 1} \frac{1 - \sqrt{2x^2 - 1}}{x - 1}$ 1.3 $\lim_{x \rightarrow 6^-} \frac{|x - 6|}{x - 6}$ 1.4 $\lim_{x \rightarrow 2} \frac{\sin(x^2 - 4)}{x - 2}$

2. Consider the function:

$$f(x) = \begin{cases} \frac{\sin 2x}{x} & \text{if } x \neq 0 \\ \ln a & \text{if } x = 0 \end{cases}$$

For which value of a will f be continuous at $x = 0$?

(a)	$a = 0$	(b)	$a = 1$	(c)	$a = e^2$	(d)	$a = 2$	(e)	$a = e$
(f)	None of these								

D. YOUTUBE VIDEOS

Kahn Academy Continuity at a point.

E. SELECTED ANSWERS

A p 102 no 21. $\frac{1}{6}$ 23. $-\frac{1}{9}$ 39. Squeeze Theorem, 0 45. $-\infty$ 49 (a) (i) 5 (ii) -5 (b) No, does not exist

p 196 39. $\frac{5}{3}$ 45. $\frac{1}{2}$ 50. $\frac{1}{2}$ p 124 19. Does not exist 23. $f(2) = 3$ 41. f is discont in $x = -1$, f is cont in $x = 1$

45. $c = \frac{2}{3}$

B. 1. (i) $k = 1$ (ii) $k = 2$ (iii) f cont for no value of k

2. $\lim_{x \rightarrow 0^-} f(x) = 0$, $\lim_{x \rightarrow 0^+} f(x) = \infty$, so f is discont in $x = 0$ 3(i) $[-1, 1]$ (ii) $(-1, 1)$

C. 1.1 -2 1.2 -2 1.3 -1 1.4 4 2. (c)

A. TEXTBOOK PROBLEMS

p 102

$$\begin{aligned} \text{no 21. } \lim_{h \rightarrow 0} \frac{\sqrt{9+h} - 3}{h} &\times \frac{\sqrt{9+h} + 3}{\sqrt{9+h} + 3} \\ &= \lim_{h \rightarrow 0} \frac{9+h-9}{h(\sqrt{9+h}+3)} \\ &= \lim_{h \rightarrow 0} \frac{1}{\sqrt{9+h}+3} = \frac{1}{6} \end{aligned}$$

$$\begin{aligned} 23. \lim_{x \rightarrow 3} \frac{\frac{1}{x} - \frac{1}{3}}{x-3} \\ &= \lim_{x \rightarrow 3} \frac{\frac{3-x}{3x}}{x-3} \\ &= \lim_{x \rightarrow 3} \frac{-1}{3x} = -\frac{1}{9} \end{aligned}$$

$$\begin{aligned} 39. \lim_{x \rightarrow 0} x^4 \cos \frac{2}{x} \quad (\text{For using the Squeeze Theorem}) \\ -1 \leq \cos \frac{2}{x} \leq 1 \\ \Rightarrow -x^4 \leq x^4 \cos \frac{2}{x} \leq x^4 \\ \text{But } \lim_{x \rightarrow 0} x^4 = \lim_{x \rightarrow 0} (-x^4) = 0 \\ \Rightarrow \lim_{x \rightarrow 0} x^4 \cos \frac{2}{x} = 0. \end{aligned}$$

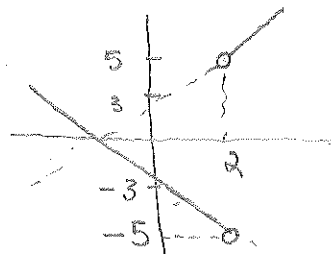
$$\begin{aligned} 45. \lim_{x \rightarrow 0^-} \left(\frac{1}{x} - \frac{1}{|x|} \right) \\ &= \lim_{x \rightarrow 0^-} \left(\frac{1}{x} + \frac{1}{x} \right) \\ &= \lim_{x \rightarrow 0^-} \left(\frac{2}{x} \right) \\ &= -\infty \end{aligned}$$

$$\begin{aligned} 49 \text{ a(i)} \lim_{x \rightarrow 2^+} g(x) &= \lim_{x \rightarrow 2^+} \frac{(x+3)(x-2)}{|x-2|} \\ &= \lim_{x \rightarrow 2^+} \frac{(x+3)(x-2)}{(x-2)} \\ &= 5 \end{aligned}$$

$$\begin{aligned} \text{(ii)} \lim_{x \rightarrow 2^-} g(x) &= \lim_{x \rightarrow 2^-} \frac{(x+3)(x-2)}{-(x-2)} \\ &= -5 \end{aligned}$$

b. No it does not exist.

c.



p 196

$$\begin{aligned} 39. \lim_{x \rightarrow 0} \frac{\sin 5x}{3x} \\ &= \frac{5}{3} \lim_{x \rightarrow 0} \frac{\sin 5x}{5x} \\ &= \frac{5}{3} \end{aligned}$$

$$\begin{aligned} 45. \lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta + \tan \theta} \\ &= \lim_{\theta \rightarrow 0} \frac{1}{\frac{\theta}{\sin \theta} + \frac{1}{\cos \theta}} \\ &= \frac{1}{2} \end{aligned}$$

$$\begin{aligned} 50. \lim_{x \rightarrow 1} \frac{\sin(x-1)}{(x+2)(x-1)} \\ &= \lim_{x-1 \rightarrow 0} \frac{\sin(x-1)}{x-1} \times \lim_{x \rightarrow 1} \frac{1}{x+2} \\ &= \frac{1}{2} \end{aligned}$$

p 124

$$\begin{aligned} 19. \lim_{x \rightarrow -1^-} (x+3) &= 2 \\ \lim_{x \rightarrow -1^+} (2^x) &= \frac{1}{2} \\ \Rightarrow \lim_{x \rightarrow -1} f(x) &\text{ does not exist.} \end{aligned}$$

$$23. \lim_{x \rightarrow 2} \frac{(x-2)(x+1)}{x-2} = 3$$

For $f(2)=3$ the function is continuous.

41. At $x = -1$

$$\lim_{x \rightarrow -1^-} f(x) = \lim_{x \rightarrow -1^-} (x^2) = 1$$

$$\lim_{x \rightarrow -1^+} f(x) = \lim_{x \rightarrow -1^+} x = -1$$

$\Rightarrow f$ is discont in $x = -1$

At $x = 1$

$$\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^-} x = 1$$

$$\lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^+} \frac{1}{x} = 1$$

$$f(1) = 1$$

$\Rightarrow f$ is cont in $x = 1$

45. $\lim_{x \rightarrow 2^-} (cx^2 + 2x) = 4c + 4$

$$\lim_{x \rightarrow 2^+} (x^3 - cx) = 8 - 3c$$

$$\Rightarrow 4c + 4 = 8 - 3c$$

$$\Rightarrow 6c = 4 \Rightarrow c = \frac{2}{3}$$

B MORE PROBLEMS

1. (i) $\lim_{x \rightarrow 1^-} (e^{-x+1}) = 1$

$\Rightarrow$ For $k=1$, f is left cont

(ii) $\lim_{x \rightarrow 1^+} \left(\frac{2}{x}\right) = 2$

$\Rightarrow$ For $k=2$, f is right cont

(iii) f is discont for any value of k because

$$\lim_{x \rightarrow 1^-} f(x) \neq \lim_{x \rightarrow 1^+} f(x)$$

* $\sqrt{1-x^2}$ cont on $[-1, 1]$ but
 $\frac{1}{\sqrt{1-x^2}}$ cont on $(-1, 1)$
 $\tan x$ cont on $(-\pi/2, \pi/2)$ (and elsewhere)

2. $\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} \frac{\cos x - 1}{\sin x} \left(\frac{0}{0}\right)$

$$= \lim_{x \rightarrow 0^-} \frac{\cos x - 1}{\sin x} \times \frac{\cos x + 1}{\cos x + 1}$$

$$= \lim_{x \rightarrow 0} \frac{\cos^2 x - 1}{\sin x}$$

$$= \lim_{x \rightarrow 0} \frac{-\sin^2 x}{\sin x}$$

$$= 0$$

$$\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} \frac{1}{\ln(x+1)} = \infty$$

$\Rightarrow f$ is discont in $x=0$; VA $x=0$.
 3. (i) $[-1, 1]$ (ii) $(-1, 1)^*$ see below

C TEST QUESTIONS

1.1 -2

1.2 $\lim_{x \rightarrow 1} \frac{1 - \sqrt{2x^2 - 1}}{x - 1} \times \frac{1 + \sqrt{2x^2 - 1}}{1 + \sqrt{2x^2 - 1}} \left(\frac{0}{0}\right)$

$$= \lim_{x \rightarrow 1} \frac{1 - (2x^2 - 1)}{(x - 1)(1 + \sqrt{2x^2 - 1})}$$

$$= \lim_{x \rightarrow 1} \frac{2(1 - x)(1 + x)}{(x - 1)(1 + \sqrt{2x^2 - 1})}$$

$$= \frac{-4}{2} = -2$$

1.3 $\lim_{x \rightarrow 6^-} \frac{1}{x - 6}$

$$= \lim_{x \rightarrow 6^-} \frac{-(x - 6)}{x - 6}$$

$$= -1$$

1.4 $\lim_{x \rightarrow 2} \frac{\sin(x^2 - 4)}{(x - 2)(x + 2)} \cdot (x + 2)$

$$= \lim_{x \rightarrow 2} \frac{\sin(x^2 - 4)}{x^2 - 4} \cdot \lim_{x \rightarrow 2} (x + 2)$$

$$= 4$$

2. $\ln a = 2 \Rightarrow a = e^2$

(c)